

SOHAM GADGIL

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RESEARCH INTERESTS

Multi-modal foundation models, LLM Alignment, Interpretability, AI for healthcare

EDUCATION

University of Washington	Seattle, WA
▪ Ph.D. in Computer Science and Engineering	Sept. 2022 – Present (<i>Expected Graduation: June 2027</i>)
Stanford University	Stanford, CA
▪ M.S. in Computer Science GPA: 4.056	<i>Graduated: June 2021</i>
▪ Coursework: Deep Learning, Natural Language Processing, Computer Vision	
Georgia Institute of Technology	Atlanta, GA
▪ B.S. in Computer Engineering, Minor in CS GPA: 4.0 (Faculty Honors)	<i>Graduated: May 2019</i>

SELECTED PUBLICATIONS

- [C] Where to Steer: Input-Dependent Layer Selection for Steering Improves LLM Alignment [\[Paper\]](#) [\[Code\]](#)
Soham Gadgil*, Chris Lin*, Su-In Lee
EMNLP 2026 (Main Conference)
- [C] SurrogateSHAP: Training-Free Contributor Attribution for Text-to-Image (T2I) Models [\[Paper\]](#) [\[Code\]](#)
Mingyu Lu, **Soham Gadgil**, Chris Lin, Chanwoo Kim, Su-In Lee
ICML 2026
- [C] Ensembling Sparse Autoencoders [\[Paper\]](#) [\[Code\]](#)
Soham Gadgil*, Chris Lin*, Su-In Lee
ICML 2026
- [J] Transparency of medical artificial intelligence systems [\[Paper\]](#) [\[News\]](#)
Chanwoo Kim*, **Soham Gadgil***, Su-In Lee
Nature Reviews Bioengineering 2025
- [C] A framework for discovering mechanisms underlying AI prediction of protected attributes [\[Paper\]](#) [\[Code\]](#)
Soham Gadgil, [...], Su-In Lee, Roxana Daneshjou
CVPR 2024 DCAMI Workshop (Oral, Best Paper Runner-Up)
- [C] Estimating Conditional Mutual Information for Dynamic Feature Selection [\[Paper\]](#) [\[Code\]](#)
Soham Gadgil*, Ian Covert*, Su-In Lee
ICLR 2024
- [J] Transparent medical image AI via an image-text foundation model grounded in medical literature [\[Paper\]](#) [\[Code\]](#)
Chanwoo Kim, **Soham Gadgil**, Alex J. DeGrave, Zhou Ran Cai, Roxana Daneshjou, Su-In Lee
Nature Medicine 2024
- [C] Transformer-based Time-Series Biomarker Discovery for COPD Diagnosis [\[Paper\]](#)
Soham Gadgil, Joshua Galanter, Mohammadreza Negahdar
NeurIPS 2024 Workshop on Time Series in the Age of Large Models
- [C] Data Alignment for Zero-Shot Concept Generation in Dermatology AI [\[Paper\]](#)
Soham Gadgil*, Mahtab Bigverdi*
ICLR 2024 Workshop on Navigating and Addressing Data Problems for Foundation Model
- [C] Combining Expert Annotations with DNN-generated Saliency Maps for X-ray Segmentation [\[Paper\]](#) [\[Code\]](#)
Soham Gadgil*, Mark Endo*, Emily Wen*, Andrew Y. Ng, Pranav Rajpurkar
MIDL 2021
- [C] Spatio-Temporal Graph Convolution for Functional MRI Analysis [\[Paper\]](#) [\[Code\]](#)
Soham Gadgil, Qingyu Zhao, Adolf Pfefferbaum, Edith V. Sullivan, Ehsan Adeli, Kilina M. Pohl
MICCAI 2020

*: equal contribution [J]: journal, [C]: conference, [P]: preprint

WORK EXPERIENCE

Google – Research Intern	June 2026 – Sep 2026
▪ Working in the Health4Gemini team on evaluating health-related losses in long multi-turn Gemini conversations	
▪ Developing dynamic user simulators to scale loss discovery, identify behavioral patterns that lead to harmful responses, and generate data for hill climbing with SFT	
▪ The simulator grounded on real Gemini logs can surface at least one problematic turn in 74% of the conversations	
Genentech – Research Intern	June 2024 – Sep 2024
▪ Designed efficient transformer-based models for raw spirogram time-series data, enabling accurate prognosis of COPD exacerbations under clinical compute constraints	

- Applied patch-based BERT-style architectures with attention to model compactness and scalability, achieving SoTA across three endpoints (*NeurIPS 2024 TSALM workshop*)

Microsoft – Data Engineer

Jul 2021 – August 2022

- Worked on a small, fast-paced team to provide automation tooling for Windows experience
- Served ~**1500** customers with a RESTful smart service to spin up secure cloud VMs for OS development
- Saved over **300** hours of developer time spent on setting up machines, access policies, and repo cloning

Microsoft – Software Engineering Intern

June 2020 – Sep 2020

- Worked in the Windows Toolkits team on an Azure hosted web portal to automate backporting bug fixes
- Developed a RESTful web API using .NET Core 3.1 with a Model-View-Controller (MVC) design pattern
- Automated web-app deployments using custom CI/CD pipelines and improved team productivity by **15%**

RESEARCH EXPERIENCE

Lee Lab of AI for bioMedical Sciences (AIMS) at UW – Research Assistant

Sept 2022 - Present

Advised by Dr. Su-In Lee

- Explainable AI for LLMs
 - Developed efficient representation learning methods for LLM interpretation (*ICML 2026*)
 - Worked on aligning LLMs to desirable behaviors using post-hoc steering techniques (*EMNLP 2026*)
 - Formulated DIME, a novel information-theoretic approach for dynamic feature selection to make accurate and interpretable predictions from a subset of features (*ICLR 2024*)
- AI in Dermatology
 - Developing unsupervised concept discovery methods for medical foundation models
 - Developed a vision-language model (MONET) to enhance the transparency of AI systems in medical imaging (*Nature Medicine 2024*)
 - Used counterfactual generative models to analyse AI-specific signals enabling classifiers to detect protected attributes with high performance (*CVPR 2024 DCAMI Workshop*)

Stanford Computational Neuroscience Lab (CNS^{LAB}) – Research Assistant

Sept 2019 – March 2020

Advised by Dr. Kilian Pohl

- Used deep learning techniques to perform sex classification from functional-MRI scans
- Formulated the non-stationary nature of functional connectivity within the context of spatio-temporal graphs
- The model beat previous approaches with an accuracy of **83.7%** (*MICCAI 2020*)

Stanford Machine Learning Group (AI for Healthcare) – Research Assistant

Sep 2020 – June 2021

Advised by Dr. Andrew Ng and Dr. Pranav Rajpurkar

- Developed CheXseg, a semi-supervised method for multi-pathology segmentation
- CheXseg leverages expert annotations and saliency maps generated by image classification models
- Compared to weak supervision, CheXseg reduces the mIoU gap with radiologists by **71.6%** (*MIDL 2021*)

TEACHING

School of Computer Science and Engineering at University of Washington

Jan 2021 – June 2021

- Introduction to AI (CSE 473), Explainable AI (CSE 599)

School of Computer Science at Stanford

Jan 2021 – June 2021

- Trustworthy ML (CS 329T): TA for a new course being offered for the first time with ~50 students

Persolv.ai

Oct 2021 – Aug 2023

- Part-time instructor at a high school bootcamp teaching AI fundamentals

School of Mathematics and ECE at Georgia Tech

Aug 2017 – May 2019

- Linear Algebra (Math 1554), Computer Architecture (ECE 3056), Hardware/Software Systems (ECE 2035)

PROFESSIONAL SERVICES

- Reviewer for Nature Journal, COLM 2026, ICML 2026, ICLR 2025, ICML 2025, NeurIPS 2024, MIDL 2022
- Organizing the [ICBINB-BIO Workshop](#) at NeurIPS 2026

HONORS & AWARDS

- Best paper runner-up award at CVPR 2024 DCAMI Workshop
- Recipient of the Electrical and Computer Engineering (ECE) Senior award for highest academic average
- Part of the team that developed and first offered the Trustworthy Machine Learning (CS 329T) course at Stanford
- Silver Reviewer Award at ICML 2026

SKILLS

- **Languages:** Python, Java, C++, C#, MATLAB, T-SQL, PL/SQL, HTML, CSS, JavaScript
- **Frameworks:** PyTorch, NumPy, Pandas, Scikit-Learn, Hugging Face, Tensorflow, Slurm